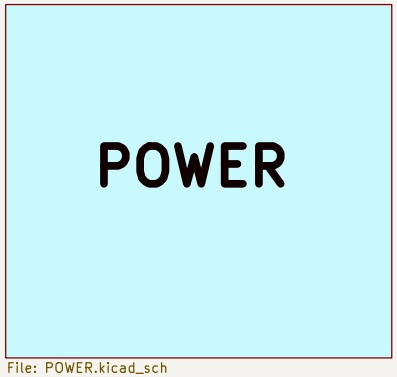
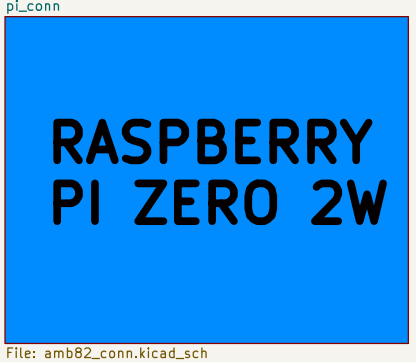


File: POWER.kicad_sch

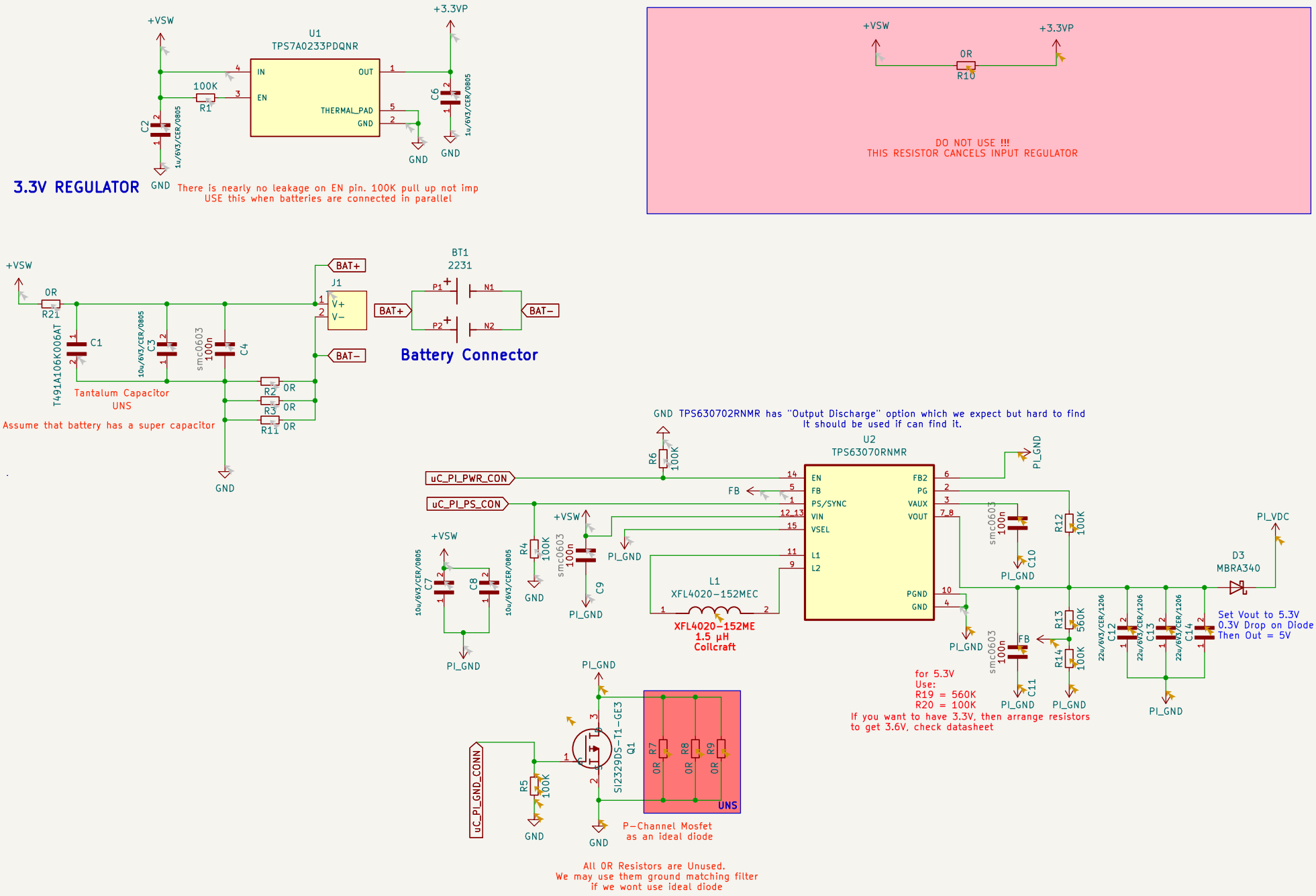


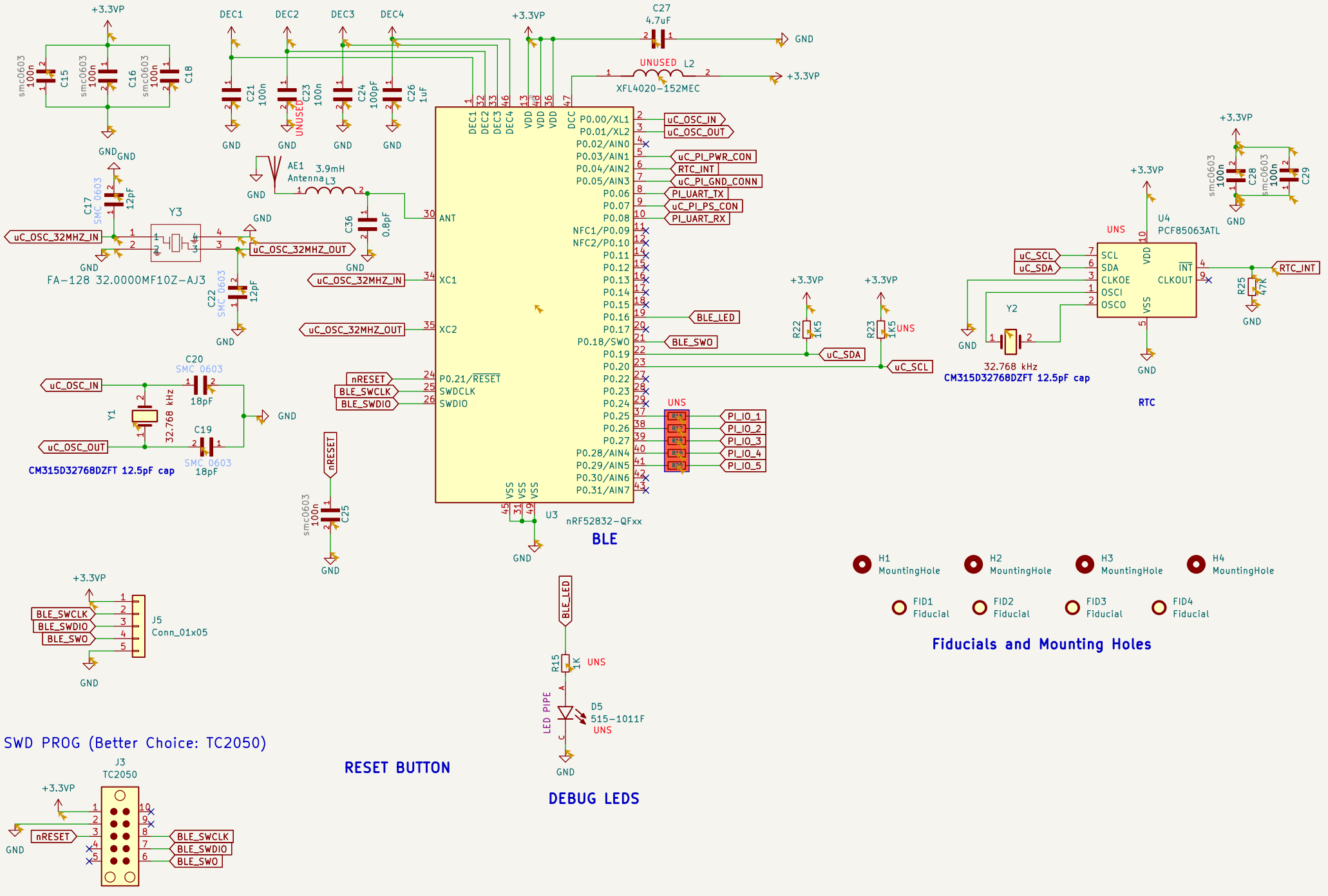
File: CPU.kicad_sch

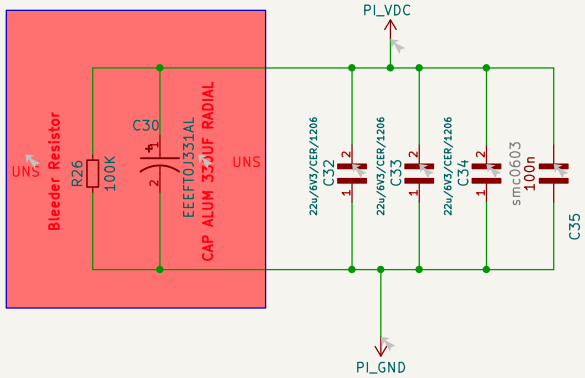
pi_conn

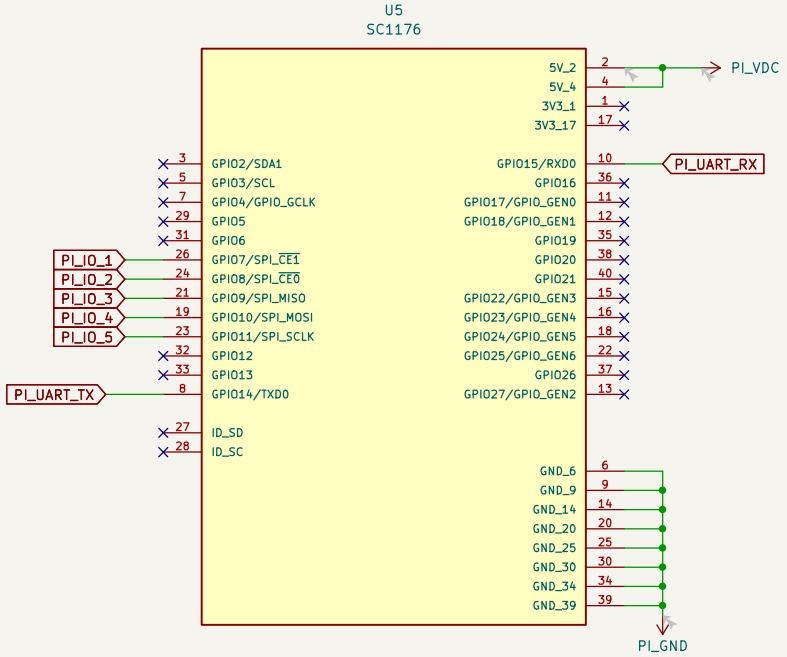


File: amb82_conn.kicad_sch


+VSW
U1
TPS7A0233PDQNR
+3.3VP
100K
R1
C2
1u/6V3/CER/0805
IN
OUT
EN
THERMAL_PAD
GND
C6
1u/6V3/CER/0805
GND
3.3V REGULATOR
GND
There is nearly no leakage on EN pin. 100K pull up not imp
USE this when batteries are connected in parallel
+VSW
0R
R10
+3.3VP
DO NOT USE !!!
THIS RESISTOR CANCELS INPUT REGULATOR
+VSW
0R
R21
C1
T491A106K006AT
Tantalum Capacitor
UNS
Assume that battery has a super capacitor
C3
10u/6V3/CER/0805
C4
smc0603
100n
R2
0R
R3
0R
R11
0R
GND
BAT+
J1
V+
V−
BAT−
BT1
2231
P1
N1
P2
N2
BAT+
BAT−
Battery Connector
GND TPS630702RNMR has "Output Discharge" option which we expect but hard to find
It should be used if can find it.
U2
TPS63070RNMR
R6
100K
uC_PI_PWR_CON
uC_PI_PS_CON
FB
EN
FB
PS/SYNC
VIN
VSEL
L1
L2
PGND
GND
FB2
PG
VAUX
VOUT
PI_GND
+VSW
C7
10u/6V3/CER/0805
C8
10u/6V3/CER/0805
R4
100K
smc0603
100n
C9
PI_GND
L1
XFL4020−152MEC
XFL4020−152ME
1.5 µH
Coilcraft
smc0603
100n
C10
R12
100K
PI_VDC
D3
MBRA340
Set Vout to 5.3V
0.3V Drop on Diode
Then Out = 5V
R13
560K
R14
100K
FB
C11
C12
22u/6V3/CER/1206
C13
22u/6V3/CER/1206
C14
22u/6V3/CER/1206
for 5.3V
Use:
R19 = 560K
R20 = 100K
If you want to have 3.3V, then arrange resistors
to get 3.6V, check datasheet
PI_GND
uC_PI_GND_CONN
R5
100K
Q1
SI2329DS-T1-GE3
R7
0R
R8
0R
R9
0R
UNS
P−Channel Mosfet
as an ideal diode
All 0R Resistors are Unused.
We may use them ground matching filter
if we wont use ideal diode


+3.3VP
C15
C16
C18
C21
C23
C24
C26
C27
4.7uF
UNUSED
L2
XFL4020-152MEC
DEC1
DEC2
DEC3
DEC4
GND
AE1
Antenna
3.9mH
L3
C17
12pF
Y3
FA-128 32.0000MF10Z-AJ3
uC_OSC_32MHZ_IN
uC_OSC_32MHZ_OUT
C22
C36
0.8pF
C20
18pF
Y1
32.768 kHz
C19
CM315D32768DZFT 12.5pF cap
nRESET
BLE_SWCLK
BLE_SWDIO
C25
ANT
XC1
XC2
P0.21/RESET
SWDCLK
SWDIO
P0.00/XL1
P0.01/XL2
P0.02/AIN0
P0.03/AIN1
P0.04/AIN2
P0.05/AIN3
P0.06
P0.07
P0.08
NFC1/P0.09
NFC2/P0.10
P0.11
P0.12
P0.13
P0.14
P0.15
P0.16
P0.17
P0.18/SWO
P0.19
P0.20
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
P0.28/AIN4
P0.29/AIN5
P0.30/AIN6
P0.31/AIN7
uC_OSC_IN
uC_OSC_OUT
uC_PI_PWR_CON
RTC_INT
uC_PI_GND_CONN
PI_UART_TX
uC_PI_PS_CON
PI_UART_RX
BLE_LED
BLE_SWO
uC_SDA
uC_SCL
PI_IO_1
PI_IO_2
PI_IO_3
PI_IO_4
PI_IO_5
UNS
R22
1K5
R23
U3
nRF52832-QFxx
BLE
U4
PCF85063ATL
SCL
SDA
CLKOE
OSCI
OSCO
CLKOUT
INT
VDD
VSS
R25
47K
Y2
32.768 kHz
CM315D32768DZFT 12.5pF cap
RTC
J5
Conn_01x05
SWD PROG (Better Choice: TC2050)
J3
TC2050
RESET BUTTON
R15
1K
D5
515-1011F
LED PIPE
DEBUG LEDS
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
FID4
Fiducial
Fiducials and Mounting Holes





Use a paralel resistance to charge and discharge capacitors safely
!!
Observe transient with an oscilloscope
and set R and C to get a proper time constant.
MAY BE EEUFS0J472B and a Bleeder Resistor
!!

If we use TPS630702RNMR then we will not need that resistor as it has
Quick Output Discharge (QOD) ability